

Pavlo Fedyna

Senior Software Engineer

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Summary

Software engineer with 15+ years of experience building scalable, high-traffic services and data processing pipelines on AWS — backend-leaning, comfortable across the stack.

Currently focused on search systems, hybrid/semantic search, and LLM-powered conversational interfaces at one of Germany's largest retailers.

Core stack spans Java, Rust, Kotlin and Python on AWS, with a growing focus on Rust for performance-critical services.

XP practitioner who ties technical decisions to customer and business outcomes.

Experience

Senior Software Engineer at Otto&Co

Oct 2021 - Present

Architecting and developing the search platform for Germany's largest online retailer.

Developed an agentic conversational search assistant (storefront, pre-sales AI chatbot) and an LLM-backed Query Intent service.

Prototyped voice-to-voice conversational search.

Growing and maintaining the Product Display Page that serves ~50% of the shop traffic.

Bringing features live via an extensive suite of A/B tests.

Built and maintained a hybrid search core service combining lexical and semantic search;

rewrote it in Rust, improving performance and cutting instance count 4x.

Maintaining search retrieval via a high-performance Solr cluster and its high-throughput index ingestion pipeline.

Own the observability and resilience stack; optimized AWS runtime costs.

Migrated infrastructure from CloudFormation to Terraform and CI/CD from Jenkins to Github Actions.

Mentoring junior developers and working students.

Java · Kotlin · Rust · Python · Go · TypeScript · Spring · Spring WebFlux · Helidon 4 · Svelte · Solr · Kafka · Hazelcast · AWS · AWS CDK · Docker · Terraform · Github Actions · Jenkins · Packer · Grafana · Prometheus · Mimir · Langchain4j

Tech Lead at Caregonexx

Oct 2020 - Sep 2021

Led a team of 5 engineers, collaborating with the PO on quarterly OKRs.

Refined user and system requirements before feeding them into the development pipeline.

Designed and developed a SaaS solution for fleet and shipment management (Java/Spring backend, React frontend).

Built a tour cost and time optimization service.

Java · TypeScript · SQL · Spring · React · AWS · Docker · PostgreSQL

Senior Software Engineer at Caregonexx

Aug 2020 - Oct 2020

Implemented and maintained an internal customer-facing solution for fleet and shipment management.
Integrated GPS providers for real-time fleet tracking.
Introduced TDD, pair programming, and CI/CD practices to the team.

Java · TypeScript · SQL · Spring · React · AWS · Docker · PostgreSQL

Senior Software Engineer at Zalando Marketing Services

Feb 2018 - Aug 2020

Built the reporting solution for Zalando's self-service B2B SaaS platform for ad management.
Rewrote the reporting and analytics pipeline, migrating to cloud and achieving significant reduction in daily operational costs (10x).
Implemented high-load data stream consumers powering the analytical processing pipeline.
Developed an analytics module for the marketplace platform.
Established 24/7 on-call support infrastructure for the team.
Introduced TDD, pair programming, and CI/CD practices.
Enabled stakeholder access to data through self-service sharing solutions.

Python · Java · Scala · SQL · Spring · Spark · Airflow · Databricks · AWS · Docker · Kubernetes · PostgreSQL · GraphQL · React · Jenkins

Senior Software Engineer at EOS Technology Solutions

Aug 2017 - Feb 2018

Implemented and maintained a client-data management system.
Refactored and added test coverage to legacy parts of the system.
Optimized the ORM layer to improve user experience.
Introduced BDD and automated web testing.

Java · SQL · JavaScript · Spring · Hibernate · React · Selenium · Cucumber · Kafka · Jenkins

Lead Software Engineer at Smaato Inc.

Mar 2017 - Aug 2017

Led development of the B2B SaaS client portal, delivering the MVP on time despite staffing constraints and high management pressure.
Led development of the ad delivery auction management system.
Built a plugin framework for the auction system, enabling sales engineers to onboard advertisers without developer involvement.

Java · SQL · JavaScript · Python · React · Selenium · Docker · AWS · Jenkins

Software Engineer at Smaato Inc.

Oct 2014 - Mar 2017

Implemented and maintained a high-load, distributed ad delivery auction management system.
Refactored the auction handling algorithm, improving CPU performance by 20% and reducing cluster size.
Migrated and rewrote several legacy applications and components.

Java · SQL · JavaScript · Spring · Hibernate · Docker · AWS · Jenkins · Packer · Kafka

Software Engineer at Edvantis Inc.

Mar 2012 - Sep 2014

Delivered a variety of projects for US and European customers.

Java · SQL · JavaScript · Spring · AWS

Junior Software Engineer at Lvivsoft Inc.

Jul 2011 - Mar 2012

Implemented and maintained a social media platform.
Deployed and tested application backend and frontend.

Java · SQL · JavaScript · Spring · Solr · AWS

Skills

Languages: Java, Kotlin, Python, Rust, TypeScript, JavaScript, SQL
Frameworks: Spring, Spring WebFlux, Helidon 4, React, GraphQL, Langchain4j
Infrastructure: AWS, AWS CDK, Docker, Kubernetes, Terraform, Kafka, Hazelcast, Github Actions, Jenkins, Packer
Observability: Grafana, Prometheus, Mimir
Data: Solr, PostgreSQL, Spark, Airflow, Databricks

Education

B.S in Applied Linguistics
State University 'Lviv Polytechnics'

Aug 2006 - Aug 2010

Languages

English, German, Ukrainian

Interests

Rust on Embedded Systems — Exploring bare-metal programming and IoT applications with Rust's safety guarantees

Game Development — Building games as a creative outlet and to explore real-time systems architecture

Homelab Infrastructure — Running a self-hosted Proxmox cluster with Kubernetes for learning and experimentation

Work Preferences

Employment: Full-time
Arrangements: Hybrid, Remote (preferred)
Travel: Open to limited travel as needed